fn match_num_groups_predicate

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `match_num_groups_predicate` in `mod.rs` at commit f5bb719 (outdated[1]).

1 Hoare Triple

Precondition

Compiler-verified

Types matching pseudocode.

Caller-verified

None

Pseudocode

```
def match_num_groups_predicate(
    ranks: Expr,
    partition_by: Vec[Expr],
    identifier: Expr,
    threshold: u32,
) -> Optional[Bound]:
    # check if is a rank function
    if not isinstance(ranks, Expr.Function) or not isinstance(
        ranks.function, FunctionExpr.Rank
    ):  #
        return None

    input = ranks.input
    options = ranks.function.options

    if partition_by != [identifier]:  #
        raise "num_groups truncation must use the identifier in the over clause"

    if not isinstance(options.method, RankMethod.Dense):  #
        raise "num_groups truncation's rank must be dense"

    if len(input) != 1:  #
        raise "rank function must be applied to a single input"

```

[1] See new changes with `git diff f5bb719..ac89f72cd rust/src/transformations/make_stable_lazyframe/truncate/matching/mod.rs`

```
25      input_item = input[0]
26
27      def strip_alias(expr):
28          while isinstance(expr, Expr.Alias):
29              expr = expr.expr
30          return expr
31
32      # Treat as_struct as a special case that represents multiple columns.
33      if isinstance(input_item, Expr.Function) and isinstance(
34          input_item.function, FunctionExpr.AsStruct
35      ):  #
36          fields = input_item.input
37          # Hash and key fields may have distinct aliases to avoid name collisions.
38          if fields:
39              first = strip_alias(fields[0])
40              second = strip_alias(fields[1]) if len(fields) > 1 else None
41              if isinstance(first, Expr.Function) and isinstance(
42                  first.function, FunctionExpr.Hash
43              ):  #
44                  if first.input[0] == second:
45                      if not isinstance(second, Expr.Function) or not isinstance(
46                          second.function, FunctionExpr.AsStruct
47                      ):
48                          raise f"expected hash input to be a struct, found {first.input}"
49                      fields = second.input
50          by = set(fields)
51      else:
52          by = {input_item}
53
54      return Bound(by=by, per_group=None, num_groups=threshold)  #
```

Postcondition

Theorem 1.1 (Postcondition)**.** If `ranks` is a dense ranking of grouping columns, and `partition_by` is a singleton of `identifier`, then returns the bound on per-identifier contributions, or an error if the truncation is mis-specified.

Proof. Due to the ambiguity between matching predicates that bound `num_groups` or `per_group`, an error is only raised if the predicate is unambiguously a `per_group` truncation predicate.

The `per_group` predicate is only unambiguously identified. This check happens on line 10.

Line 16 checks that `partition_by` is a singleton of the `identifier`, meeting the conditions of the postcondition.

We now check whether the truncation predicate is well-defined, on lines 19 and 22.

Finally, line 35 extracts the grouping columns from the predicate. Special consideration is made for structs, which are considered multiple grouping columns.

Further consideration is made if the first field is a hash of the second field, on line 43. Field aliases are ignored when checking this relationship: they prevent duplicate field names in Polars without changing the values. This deterministically orders grouping columns by hash and then by the original keys; it does not produce a uniformly random sample. The second field of grouping columns is kept in the ranker for the rare case that there are hash collisions, which prevents different combinations of grouping keys from being assigned the same rank, preventing more distinct groups being kept than is permitted.

This predicate corresponds to a `num_groups` truncation predicate, because the over expression groups by the `identifier` column, and within each group, a dense ranking is applied to unique combinations of the grouping columns. If the only rows kept are those corresponding to grouping keys assigned dense ranks at most `threshold`, then each user identifier will have at most `threshold` unique combinations of the grouping columns after filtering by the predicate.

Therefore the bound on user contributions constructed on line 54 is valid. □